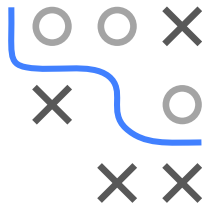


# RISK MINIMIZER AND OPTIMAL CONSTANT

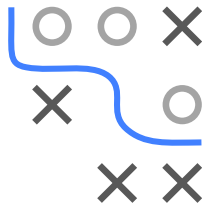
| Name                  | Risk Minimizer                                                                                                      | Optimal Constant                                           |
|-----------------------|---------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------|
| L2                    | $f^*(\mathbf{x}) = \mathbb{E}_{y \mathbf{x}} [y \mid \mathbf{x}]$                                                   | $\hat{f}(\mathbf{x}) = \frac{1}{n} \sum_{i=1}^n y^{(i)}$   |
| L1                    | $f^*(\mathbf{x}) = \text{med}_{y \mathbf{x}} [y \mid \mathbf{x}]$                                                   | $\hat{f}(\mathbf{x}) = \text{med}(y^{(i)})$                |
| 0-1                   | $h^*(\mathbf{x}) = \arg \max_{l \in \mathcal{Y}} \mathbb{P}(y = l \mid \mathbf{x})$                                 | $\hat{h}(\mathbf{x}) = \text{mode} \{y^{(i)}\}$            |
| Brier                 | $\pi^*(\mathbf{x}) = \mathbb{P}(y = 1 \mid \mathbf{x})$                                                             | $\hat{\pi}(\mathbf{x}) = \frac{1}{n} \sum_{i=1}^n y^{(i)}$ |
| Bernoulli (on probs)  | $\pi^*(\mathbf{x}) = \mathbb{P}(y = 1 \mid \mathbf{x})$                                                             | $\hat{\pi}(\mathbf{x}) = \frac{1}{n} \sum_{i=1}^n y^{(i)}$ |
| Bernoulli (on scores) | $f^*(\mathbf{x}) = \log \left( \frac{\mathbb{P}(y=1 \mid \mathbf{x})}{1 - \mathbb{P}(y=1 \mid \mathbf{x})} \right)$ | $\hat{f}(\mathbf{x}) = \log \frac{n+1}{n-1}$               |



We see: For regression, the RMs model the conditional expectation and median of the underlying distribution. This makes intuitive sense, depending on your concept of how to best estimate central location / how robust this location should be.

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| Name                  | Risk Minimizer                                                                                                      | Optimal Constant                                           |
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| L1                    | $f^*(\mathbf{x}) = \text{med}_{y \mathbf{x}} [y \mid \mathbf{x}]$                                                   | $\hat{f}(\mathbf{x}) = \text{med}(y^{(i)})$                |
| 0-1                   | $h^*(\mathbf{x}) = \arg \max_{l \in \mathcal{Y}} \mathbb{P}(y = l \mid \mathbf{x})$                                 | $\hat{h}(\mathbf{x}) = \text{mode} \{y^{(i)}\}$            |
| Brier                 | $\pi^*(\mathbf{x}) = \mathbb{P}(y = 1 \mid \mathbf{x})$                                                             | $\hat{\pi}(\mathbf{x}) = \frac{1}{n} \sum_{i=1}^n y^{(i)}$ |
| Bernoulli (on probs)  | $\pi^*(\mathbf{x}) = \mathbb{P}(y = 1 \mid \mathbf{x})$                                                             | $\hat{\pi}(\mathbf{x}) = \frac{1}{n} \sum_{i=1}^n y^{(i)}$ |
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For the 0-1 loss, the risk minimizer constructs the **optimal Bayes decision rule**: We predict the class with maximal posterior probability.

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| Name                  | Risk Minimizer                                                                                                      | Optimal Constant                                           |
|-----------------------|---------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------|
| L2                    | $f^*(\mathbf{x}) = \mathbb{E}_{y \mathbf{x}} [y \mid \mathbf{x}]$                                                   | $\hat{f}(\mathbf{x}) = \frac{1}{n} \sum_{i=1}^n y^{(i)}$   |
| L1                    | $f^*(\mathbf{x}) = \text{med}_{y \mathbf{x}} [y \mid \mathbf{x}]$                                                   | $\hat{f}(\mathbf{x}) = \text{med}(y^{(i)})$                |
| 0-1                   | $h^*(\mathbf{x}) = \arg \max_{l \in \mathcal{Y}} \mathbb{P}(y = l \mid \mathbf{x})$                                 | $\hat{h}(\mathbf{x}) = \text{mode} \{y^{(i)}\}$            |
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| Bernoulli (on probs)  | $\pi^*(\mathbf{x}) = \mathbb{P}(y = 1 \mid \mathbf{x})$                                                             | $\hat{\pi}(\mathbf{x}) = \frac{1}{n} \sum_{i=1}^n y^{(i)}$ |
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For Brier and Bernoulli, we predict the posterior probabilities (of the true DGP!). Losses that have this desirable property are called **proper scoring (rules)**.

